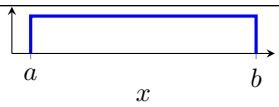
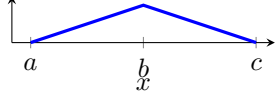
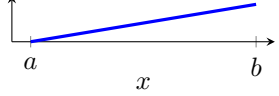
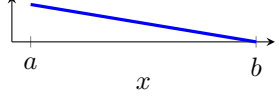
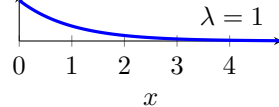
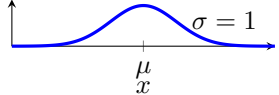


# G11 T1 C5 - Mean and Variance

| Distribution      | Notation                               | $E(X)$              | $Var(X)$                                    | Graph                                                                               |
|-------------------|----------------------------------------|---------------------|---------------------------------------------|-------------------------------------------------------------------------------------|
| Uniform           | $X \sim U(a, b)$                       | $\frac{a+b}{2}$     | $\frac{(b-a)^2}{12}$                        |  |
| Triangular        | $X \sim \text{Tri}(a, b, c)$           | $\frac{a+b+c}{3}$   | $\frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$ |  |
| Linear increasing | $X \sim \text{Lin}_{\text{inc}}(a, b)$ | $\frac{a+2b}{3}$    | $\frac{(b-a)^2}{18}$                        |  |
| Linear decreasing | $X \sim \text{Lin}_{\text{dec}}(a, b)$ | $\frac{2a+b}{3}$    | $\frac{(b-a)^2}{18}$                        |  |
| Exponential       | $X \sim \text{Exp}(\lambda)$           | $\frac{1}{\lambda}$ | $\frac{1}{\lambda^2}$                       |  |
| Normal            | $X \sim N(\mu, \sigma^2)$              | $\mu$               | $\sigma^2$                                  |  |

- Customer Spending.** A company tracks the amount spent by customers during a typical shopping visit. Based on its historical data, the spending amount is approximately centered at \$75, with a standard deviation of \$18. **Task:** Find the mean, variance, and standard deviation of the amount spent by a randomly selected customer, measured in dollars.
- Construction Repair Time.** A construction company estimates the time required to complete a small repair. Based on previous projects, the completion time ranges from 6 to 18 hours. Most repairs are completed in about 10 hours, while shorter and longer completion times are less common. Therefore, the completion time has a triangular probability density with a mode at 10 hours. **Task:** Find the mean, variance, and standard deviation of the completion time of a randomly selected repair, measured in hours.
- Package Delivery Time.** A delivery company transports packages within a city. Based on previous deliveries, the delivery time for a randomly selected package ranges from 15 to 75 minutes. Most packages are delivered relatively quickly, so shorter delivery times are more likely and the probability density decreases linearly as the delivery time increases. At the longest delivery times, the probability can be considered negligible. **Task:** Find the mean, variance, and standard deviation of the delivery time for a randomly selected package, measured in minutes.
- Online Orders.** A supermarket receives online orders throughout the day. Based on its recent activity, the supermarket receives an average of 12 orders per hour. **Task:** Find the mean, variance, and standard deviation of the time between consecutive orders, measured in minutes.
- Coffee Shop Waiting Time.** A coffee shop records the time customers wait for their orders during the morning. Based on previous observations, the waiting time ranges from 3 to 15 minutes, with all values in this range considered equally likely. **Task:** Find the mean, variance, and standard deviation of the waiting time for a randomly selected customer, measured in minutes.
- Furniture Production Time.** A manufacturer produces customized furniture. Based on previous orders, the time required to complete a randomly selected piece ranges from 4 to 12 hours. More complex pieces are more common, so longer production times are more likely and the probability density increases linearly across this range. At the shortest production times, the probability can be considered negligible. **Task:** Find the mean, variance, and standard deviation of the production time for a randomly selected piece, measured in hours.

# 1 Solutions

## C4.1 - Finding the mean, variance, and standard deviation

The amount spent by a randomly selected customer is approximately normally distributed, with mean \$75 and standard deviation \$18.

|                |                               |
|----------------|-------------------------------|
| <b>General</b> | $X \sim N(\mu, \sigma^2)$     |
| <b>Context</b> | $\mu = 75, \quad \sigma = 18$ |

Therefore,

$$E(X) = \mu = 75$$

$$Var(X) = \sigma^2 = 18^2 = 324$$

$$\sigma_X = \sigma = 18$$

Thus,

$$E(X) = 75, \quad Var(X) = 324, \quad \sigma_X = 18$$

## C4.2 - Finding the mean, variance, and standard deviation

The completion time ranges from 6 to 18 hours, with a mode at 10 hours. Therefore, the triangular distribution has

|                |                                     |
|----------------|-------------------------------------|
| <b>General</b> | $X \sim \text{Tri}(a, b, c)$        |
| <b>Context</b> | $a = 6, \quad b = 18, \quad c = 10$ |

Therefore,

$$E(T) = \frac{a + b + c}{3} = \frac{6 + 18 + 10}{3} = \frac{34}{3} \approx 11.333$$

$$Var(T) = \frac{a^2 + b^2 + c^2 - ab - ac - bc}{18}$$

$$= \frac{6^2 + 18^2 + 10^2 - (6)(18) - (6)(10) - (18)(10)}{18}$$

$$= \frac{112}{18} = \frac{56}{9} \approx 6.222$$

$$\sigma_T = \sqrt{Var(T)} = \sqrt{\frac{56}{9}} \approx 2.494$$

Thus,

$$E(T) \approx 11.333, \quad Var(T) \approx 6.222, \quad \sigma_T \approx 2.494$$

## C4.3 - Finding the mean, variance, and standard deviation

The delivery time ranges from 15 to 75 minutes, with shorter delivery times more likely and the probability density decreasing linearly as the delivery time increases.

|                |                                        |
|----------------|----------------------------------------|
| <b>General</b> | $X \sim \text{Lin}_{\text{dec}}(a, b)$ |
| <b>Context</b> | $a = 15, \quad b = 75$                 |

Therefore,

$$E(T) = \frac{2a + b}{3} = \frac{2(15) + 75}{3} = 35$$

$$Var(T) = \frac{(b - a)^2}{18} = \frac{(75 - 15)^2}{18} = \frac{3600}{18} = 200$$

$$\sigma_T = \sqrt{Var(T)} = \sqrt{200} \approx 14.142$$

Thus,

$$E(T) = 35, \quad Var(T) = 200, \quad \sigma_T \approx 14.142$$

#### C4.4 - Finding the mean, variance, and standard deviation

The supermarket receives an average of 12 orders per hour. Therefore, the rate in minutes is

$$\lambda = \frac{12}{60} = 0.20 \text{ orders/minute.}$$

Let  $X$  represent the time between consecutive orders.

|                |                              |
|----------------|------------------------------|
| <b>General</b> | $X \sim \text{Exp}(\lambda)$ |
| <b>Context</b> | $\lambda = 0.20$             |

Therefore,

$$E(X) = \frac{1}{\lambda} = \frac{1}{0.20} = 5$$

$$\text{Var}(X) = \frac{1}{\lambda^2} = \frac{1}{(0.20)^2} = 25$$

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{25} = 5$$

Thus,

$$E(X) = 5, \quad \text{Var}(X) = 25, \quad \sigma_X = 5$$

#### C4.5 - Finding the mean, variance, and standard deviation

The waiting time ranges from 3 to 15 minutes, with all values in this range considered equally likely. Therefore, the waiting time follows a uniform distribution with

|                |                       |
|----------------|-----------------------|
| <b>General</b> | $X \sim U(a, b)$      |
| <b>Context</b> | $a = 3, \quad b = 15$ |

Therefore,

$$E(X) = \frac{a+b}{2} = \frac{3+15}{2} = 9$$

$$\text{Var}(X) = \frac{(b-a)^2}{12} = \frac{(15-3)^2}{12} = \frac{144}{12} = 12$$

$$\sigma_X = \sqrt{\text{Var}(X)} = \sqrt{12} \approx 3.464$$

Thus,

$$E(X) = 9, \quad \text{Var}(X) = 12, \quad \sigma_X \approx 3.464$$

#### C4.6 - Finding the mean, variance, and standard deviation

The production time ranges from 4 to 12 hours, with longer production times more likely and the probability density increasing linearly across this range.

|                |                                        |
|----------------|----------------------------------------|
| <b>General</b> | $X \sim \text{Lin}_{\text{inc}}(a, b)$ |
| <b>Context</b> | $a = 4, \quad b = 12$                  |

Therefore,

$$E(T) = \frac{a+2b}{3} = \frac{4+2(12)}{3} = \frac{28}{3} \approx 9.333$$

$$\text{Var}(T) = \frac{(b-a)^2}{18} = \frac{(12-4)^2}{18} = \frac{64}{18} = \frac{32}{9} \approx 3.556$$

$$\sigma_T = \sqrt{\text{Var}(T)} = \sqrt{\frac{32}{9}} \approx 1.886$$

Thus,

$$E(T) \approx 9.333, \quad \text{Var}(T) \approx 3.556, \quad \sigma_T \approx 1.886$$